

Banking Analytics Case Study

Project Overview

This case study documents a **Banking Data Analysis** project completed using SQL, Python, Excel, and Power BI. The project analyzes customer, account, loan, and transaction data to understand banking operations, identify business trends, and generate meaningful insights through data analysis and interactive dashboards.

The project follows an end-to-end data analytics workflow, starting from data cleaning and preparation to analysis, visualization, and business reporting. All insights and observations presented in this case study are based on the analysis performed using the project's datasets and are intended to support data-driven decision-making.

Analysis Objective

The objective of this project was to analyze banking data to:

- Understand customer distribution across different cities, occupations, and account types.
 - Analyze account balances and identify customer segments based on banking activity.
 - Examine loan distribution across different loan types, interest rates, and loan status.
 - Study transaction patterns across various banking channels, months, and customer locations.
 - Develop interactive dashboards and KPI reports to support data-driven decision-making.
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Dataset Description

- **Total Customers:** 5,000
- **Total Accounts:** 5,000
- **Total Loans:** 5,000
- **Total Transactions:** 100,000

Key Attributes

- Customer_ID
- Customer_Name
- Account_ID
- Account_Type
- Balance
- Loan_ID

- Loan_Type
- Loan_Amount
- Interest_Rate
- Transaction_ID
- Transaction_Date
- Transaction_Channel
- Transaction_Amount
- City

The dataset consists of four interconnected tables linked through primary and foreign keys, enabling comprehensive analysis of customer behavior, account performance, loan portfolios, and transaction activities. Before analysis, the data was cleaned, validated, and prepared to ensure consistency across all datasets.

Tools & Technologies Used

- **Programming Language:** Python
 - **Libraries:**
 - **Pandas** – Data cleaning and analysis
 - **NumPy** – Numerical operations
 - **Matplotlib & Seaborn** – Data visualization
 - **Database:** MySQL – Data storage and SQL analysis
 - **SQL:** Joins, Aggregate Functions, Window Functions, CTEs
 - **Data Handling:** CSV Files
 - **Google Sheets:** Formulas, Pivot Tables, Charts
 - **Power BI:** Interactive dashboards and KPI reporting
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Data Preparation & Preprocessing

The banking datasets were prepared before analysis to ensure accurate and reliable results.

- Removed duplicate records from the datasets.
 - Checked and handled missing or invalid values.
 - Corrected data types for dates, numbers, and text fields.
 - Standardized column names and formatting across all datasets.
 - Validated relationships between Customers, Accounts, Loans, and Transactions using common keys.
 - Imported the cleaned datasets into MySQL, Excel, Python, and Power BI for analysis.
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Analysis Performed

The banking dataset was analyzed using SQL, Python, Excel, and Power BI to identify trends, measure business performance, and generate actionable insights.

1. Customer Analysis

- Analyzed customer distribution by city, occupation, and gender.
- Identified cities with the highest customer base.

2. Account Analysis

- Analyzed account types and account balances.
- Identified customers with the highest account balances.

3. Loan Analysis

- Compared loan amounts across different loan types.
- Analyzed loan status and interest rate distribution.

4. Transaction Analysis

- Analyzed transaction volume by channel and month.
- Identified the most frequently used banking channels and monthly transaction trends.

5. Dashboard & KPI Analysis

- Built KPIs and interactive dashboards using Excel and Power BI.
- Visualized key business metrics to support decision-making.

Key Observations & Business Relevance

Key Observations

- Savings accounts contributed the highest overall account balance among all account types.
- Mobile banking was the most frequently used transaction channel, reflecting strong digital banking adoption.
- Home loans represented the largest share of the bank's loan portfolio.
- Customer distribution varied across cities, with a few cities contributing a larger customer base.
- Monthly transaction analysis highlighted fluctuations in transaction values throughout the year.

Business Relevance

The analysis provides valuable insights that can support banking operations and business decision-making by:

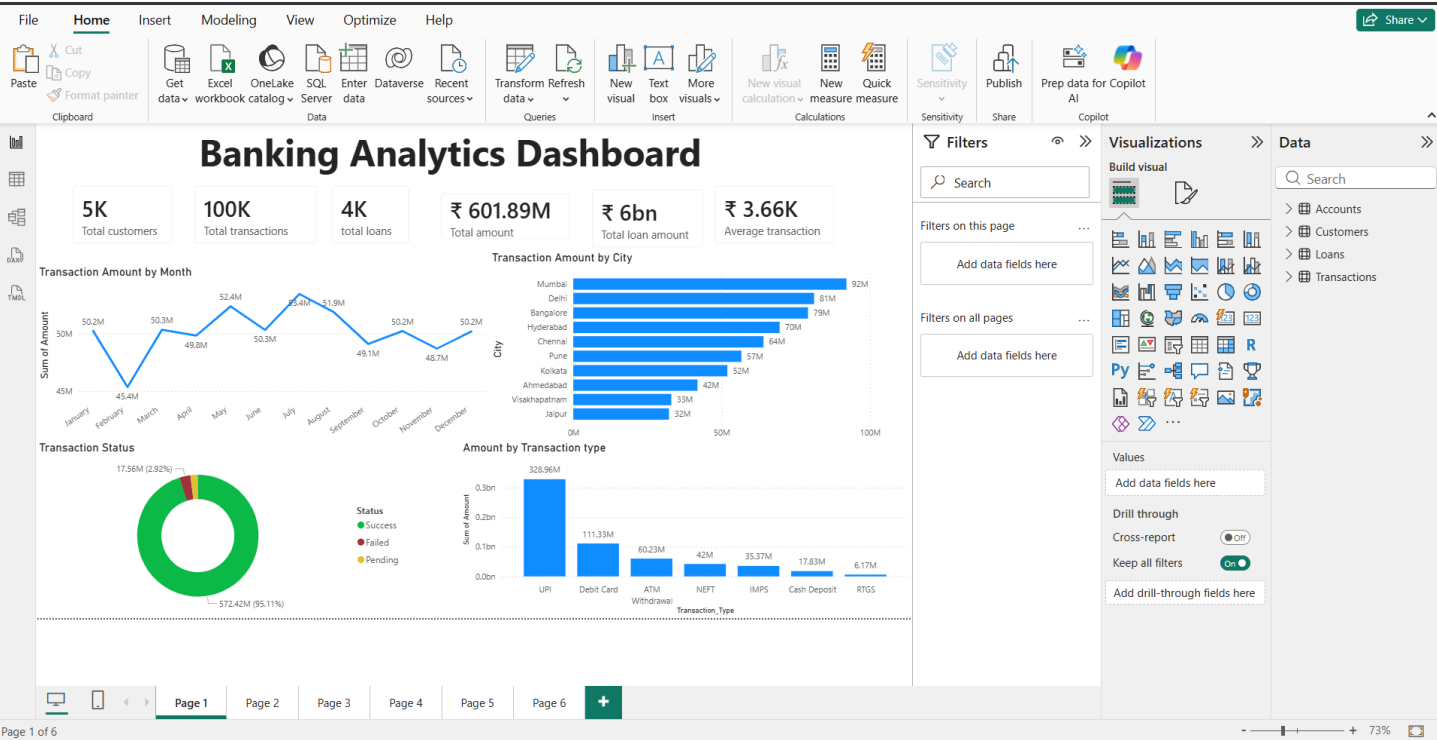
- Identifying high-value customers for premium banking services.
- Understanding customer preferences across different banking channels.
- Monitoring loan portfolio performance and customer borrowing trends.
- Supporting management with KPI-driven reports and interactive dashboards.

Resume-Ready Project Summary

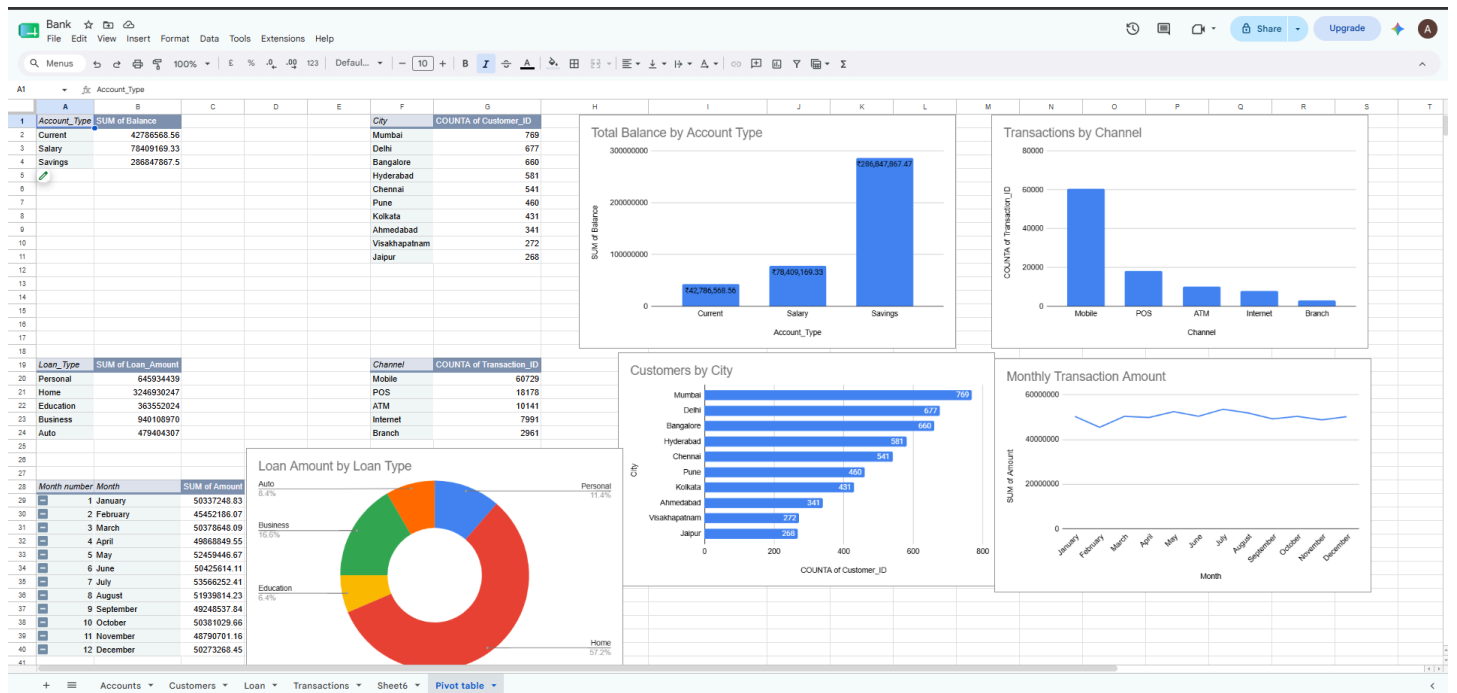
Banking Data Analysis Project | SQL, Python, Excel, Power BI

- Performed end-to-end analysis on banking data consisting of customer, account, loan, and transaction datasets.
- Used SQL to write business queries, joins, aggregate functions, and window functions for data analysis.
- Performed data cleaning, exploratory data analysis (EDA), and visualizations using Python (Pandas, NumPy, Matplotlib, and Seaborn).
- Built KPI reports, Pivot Tables, and charts in Excel to summarize business performance.
- Developed an interactive Power BI dashboard to visualize key banking metrics and support business decision-making.
- Generated actionable business insights and recommendations based on customer behavior, account performance, loan distribution, and transaction trends.

Power BI:



Sheets:



Conclusion

This Banking Data Analysis project gave me practical experience in working with real-world banking data using SQL, Python, Excel, and Power BI. Throughout the project, I cleaned and analyzed customer, account, loan, and transaction datasets to identify trends, measure business performance, and generate meaningful insights.

By combining multiple analytics tools, I was able to build interactive dashboards, create KPI reports, and present data in a way that supports business decision-making. This project strengthened my understanding of the complete data analytics workflow and improved my ability to solve business problems using data.

Overall, this project demonstrates my ability to transform raw data into actionable insights and deliver analytical solutions that can support better decision-making in the banking domain.